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A nonlinear state estimation framework for humanoid robots

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ABSTRACT

This article proposes a novel nonlinear state estimation framework for humanoid robots based on the dynamics of the center of mass (CoM) and dual-loop Kalman filter(DLKF). It effectively fuses the information from inertial measurement unit(IMU), joint encoders, and foot sensitive resistors (FSRs), and provides state estimates for various gait generation algorithms and dynamic balance controllers with CoM or divergence component of motion (DCM) feedback. Compared to the widely used linear models such as the linear inverted pendulum model (LIPM), the nonlinear dynamics of CoM effectively reduce the process error. However, the humanoid robot is a highly complex dynamic system with multiple links and joints, the dynamics of CoM are also a simplification of the whole body dynamics. As a result, it brings non-zero-mean, non-Gaussian, and correlated process error, which the conventional extend Kalman filter (EKF) cannot handle. To this end, the DLKF is adopted to compensate the process error, thus the estimator is robust to the modeling error caused by simplifications. Furthermore, the invariant extended Kalman filter (IEKF) is used for floating base kinematics estimation, and the force/torque (F/T) sensor which is difficult to integrate on smaller or cheaper robots due to the size and cost is not used in our framework. Finally, our nonlinear state estimation framework improves the accuracy of CoM and DCM estimation in a virtual environment simulation using our self-developed Defensor hydraulic humanoid robot.

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1. Introduction

Robust and stable gait generation is of vital importance for humanoid robots. Generating gait by using the whole body dynamics is impractical since humanoid robots are of high dimension, nonlinear, and hybrid. To date, various studies focusing on the dynamics of the center of mass(CoM) have been developed to tackle this problem. A variety of simplified models and gait generation algorithms based on the simplified models were proposed, such as linear inverted pendulum model(LIPM) [1–5], spring-loaded inverted pendulum(SLIP) [6–8], and the capture point(CP) or divergence component of motion(DCM) based approaches [9–11]. All these online approaches focused on the dynamics of CoM need the real-time feedback of the CoM position. A considerate number of these studies also need the CoM velocity feedback. Therefore, it is common knowledge that only by obtaining sound CoM state estimation, the robustness and stable gait generation can be guaranteed.

Thus, state estimation for humanoid robots is one of the most important components in locomotion control. Since the humanoid robot is a kind of mobile robot without a fixed base, the first step for its state estimation is the floating base kinematics estimation. To this end, Bloesch et al. [12] proposed the quaternion-based extended Kalman filter(Q-EKF), which fused joint encoders data with on-board inertial measurement unit(IMU) measurements. The Q-EKF assumed that there is no sliding between the robot's supporting foot and the ground in measurements. Rotella et al. [13] further developed this method while also considering the supporting foot orientation, thus improving flat foot legged robots estimation accuracy. In [14,15], the invariant extended Kalman filter(IEKF) was proposed, which has better convergence performance. Hartley et al. [16,17] adopted the IEKF for humanoid robots floating base kinematics estimation, and showed better convergence performance than the Q-EKF.

After floating base kinematics estimation, the CoM state should be estimated. Many studies use simplified process models since it is also difficult to adopt the whole body dynamics as the process model. Kwon et al. [18] adopted the LIPM as process model for CoM state estimation, and this idea was employed in many CoM state estimators [19–23]. Stephens [19] properly estimated CoM offset or external force, and illustrated the complexity to set them apart based on observability analysis. Xinjilefu et al. [20] used the same method as Stephens, and further applied it to modeling error compensation, fall detection, and prevention. Hashlamon et al. [21] modeled the disturbance mathematically in terms of previous CoM and zero moment point(ZMP) states. However, the LIPM based methods are limited

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in development due to the assumption of constant CoM height. They could not well estimate the height of the CoM. Besides, LIPM ignores the coupling between the sagittal plane and the lateral plane, and introduces errors of the CoM state estimation in the two planes. Moreover, the simplified models such as LIPM bring non-zero-mean, non-Gaussian, correlated error, which makes the traditional Kalman filter(KF) non-optimal. In order to handle this problem, Bae et al. [24] further developed KF to dual-loop Kalman filter(DLKF), the DLKF could work well in humanoid robots estimation with simplified models. Then, the humanoid robot state estimator based on the compliant inverted pendulum(CIP) and DLKF was proposed [25,26]. The CIP model adds a virtual spring and a damper to the traditional inverted pendulum model, which reflects the flexibility of the humanoid robot since the humanoid robot is a multi-link and lightweight dynamic system. However, it has a few shortcomings that the CIP process model. Initially, the CIP model ignores the dynamics between the CoM and ZMP, which is one of the most important dynamics in humanoid locomotion. Moreover, the CIP model ignores the influence that external forces bring to state estimation, which is crucial for humanoid robots in complex environments. Furthermore, the spring constant and damping coefficient change as the robot state changes. A great deal of tiresome and troublesome experiments are needed to acquire valid spring constant and damping coefficients in real robots. [26] used a simulator to obtain spring constant and damping coefficient, which is not realistic. Additionally, like the LIPM, the CIP model is still a type of linear simplified model, which makes artificial assumptions of humanoid robots locomotion, which introduces unignorable process error. Thus, the full nonlinear CoM dynamics should be used directly for the CoM state estimation. Piperakis et al. [27] used the full nonlinear CoM dynamics as process model based on EKF, defined CoM position, CoM velocity and external forces as states, and proved that all states are locally observable in all cases by using nonlinear observability analysis. In [28], the influence of the rate of angular momentum is considered, which improved the accuracy of CoM state estimation. The nonlinear model is full nonlinear centroidal dynamics, which makes no artificial assumptions of humanoid robots locomotion, and improves the estimate accuracy in the prediction step. However, humanoid robots are complex multi-body dynamic systems, the full nonlinear CoM dynamics are also simplifications of whole body dynamics. In addition, due to the parameter error, unmodeled drive system, and calibration error, this process model also brings non-zero-mean, non-Gaussian, correlated error that cannot be compensated by traditional EKF. Meanwhile, the complementary filter(CF) based CoM state estimator was also proposed. Bailly et al. [29] fused kinematic measurement and CoM dynamics based on CF. However, the CoM dynamics they used did not consider external force and the F/T sensor is necessary.

In this work, a nonlinear state estimation framework for humanoid robots is proposed. We effectively fuse the information from IMU, joint encoders and foot sensitive resistors(FSRs). The nonlinear CoM dynamics are adopted as process model, and the DLKF proposed in [24] is used to compensate for the process noise due to the simplification of whole body dynamics. The CoM dynamics and DLKF effectively reduce the process error in the prediction step. It is worth pointing out that, thanks to the nonlinear model in [28], the force/torque(F/T) sensor which is difficult to integrate on smaller and cheaper robots due to the size and cost [30] is not used in our framework. This means that the framework can be widely used in various humanoid robots. Compared to the existing nonlinear state estimation framework, the CoM state estimation accuracy is improved.

The remainder of this paper is organized as follows. The preliminaries are introduced in Section 2, which include the nonlinear dynamics of the humanoid robot and the dual-loop Kalman filter. Section 3 presents our nonlinear state estimation framework. Its performance is compared with the existing state estimation approaches in Section 4. Finally, Section 5 concludes the paper and suggests future work.

2. Preliminaries

2.1. Nonlinear dynamics of humanoid robots

This section describes the nonlinear dynamics of humanoid robots. Since the whole body dynamics of humanoid robots are very complicated as described in Section 1, we use the Newton–Euler equations to describe the nonlinear dynamics of humanoid robots:

$$m(\ddot{\boldsymbol{c}}+\boldsymbol{g})=\sum_{\boldsymbol{i}}\boldsymbol{f_i}+\boldsymbol{f_e} \tag{1}$$

$$m\boldsymbol{c}\times(\ddot{\boldsymbol{c}}+\boldsymbol{g})+\dot{\boldsymbol{L}}=\boldsymbol{c}\times\boldsymbol{f_e}+\sum_{\boldsymbol{i}}(\boldsymbol{p_i}-\boldsymbol{c})\times\boldsymbol{f_i} \tag{2}$$

where, $\boldsymbol{c}\in\boldsymbol{R}^3$ denotes the CoM position, $\boldsymbol{f_i}\in\boldsymbol{R}^3$ denotes the ground contact force, $\boldsymbol{f_e}\in\boldsymbol{R}^3$ is the external force, $\dot{\boldsymbol{L}}\in\boldsymbol{R}^3$ is the rate of angular momentum, $\boldsymbol{p_i}\in\boldsymbol{R}^3$ are the positions of the contact points, $\boldsymbol{g}\in\boldsymbol{R}^3$ is the acceleration of gravity, m is the overall mass, i is the index of contact points and contact forces.

In humanoid robots locomotion, ZMP is also an important physical quantity, which is defined as:

$$\boldsymbol{z}=\begin{bmatrix}\frac{\sum_i f_i^z p_i^x}{\sum_i f_i^z} & \frac{\sum_i f_i^z p_i^y}{\sum_i f_i^z} & z^z\end{bmatrix} \tag{3}$$

where the superscripts indicate the corresponding axis.

Combine Eqs. (1), (2) and (3), and expand the vector in the three directions of x, y, and z:

$$\ddot{c}^x=\frac{(c^x-z^x)\left(m\left(\ddot{c}^z+g^z\right)-f_e^z\right)-\dot{L}^y}{m\left(c^z-z^z\right)}+\frac{1}{m}f_e^x \tag{4}$$

$$\ddot{c}^y=\frac{(c^y-z^y)\left(m\left(\ddot{c}^z+g^z\right)-f_e^z\right)+\dot{L}^x}{m\left(c^z-z^z\right)}+\frac{1}{m}f_e^y \tag{5}$$

$$\ddot{c}^z=\frac{1}{m}\left(f_N+f_e^z\right)-g^z \tag{6}$$

where, $f_N\triangleq\sum_i f_i^z$ is the resultant force of ground contact forces in the z direction, which can be measured directly by FSRs.

Substituting Eq. (6) in Eqs. (4) and (5), we get the full nonlinear dynamics of humanoid robots:

$$\ddot{c}^x=\frac{c^x-z^x}{m\left(c^z-z^z\right)}f_N-\frac{\dot{L}^y}{m\left(c^z-z^z\right)}+\frac{1}{m}f_e^x, \tag{7}$$

$$\ddot{c}^y=\frac{c^y-z^y}{m\left(c^z-z^z\right)}f_N+\frac{\dot{L}^x}{m\left(c^z-z^z\right)}+\frac{1}{m}f_e^y. \tag{8}$$

Compared with the commonly used LIPM and CIP models in humanoid robots studies, the nonlinear dynamic equations shown in Eqs. (6)–(8) do not make any artificial assumptions about the motion of the CoM of the robot, as shown in Fig. 1. In Section 3.2, we will use this model as process model to estimate the CoM state.

2.2. Dual-loop Kalman filter

The nonlinear dynamics of humanoid robot in Section 2.1 simplify the whole body dynamics of the robot to nonlinear CoM dynamics. Although no artificial assumptions are made in the simplification process, there is still a big gap with the whole body dynamics of the robot, which introduces non-zero-mean, non-Gaussian, and correlated process error. This article uses DLKF, a

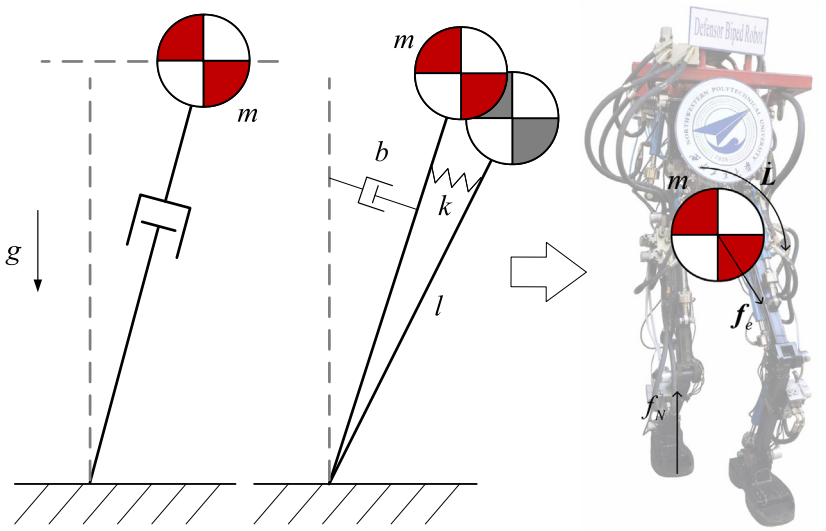


Fig. 1. LIPM(left), CIP(middle), Nonlinear dynamics(right).

Table 1
Mathematical notations for the DLKF.

Mathematical notation	Illustration
k	Time step
$\boldsymbol{x} \in \boldsymbol{R}^9$	System state
$\boldsymbol{u} \in \boldsymbol{R}^6$	System inputs
$\boldsymbol{y} \in \boldsymbol{R}^6$	Measurements
$\boldsymbol{w} \in \boldsymbol{R}^9$, $\boldsymbol{v} \in \boldsymbol{R}^6$	Process error, measurement noise
$f(\cdot)$, $h(\cdot)$	State transition function, measurement function
$\boldsymbol{A} \in \boldsymbol{R}^{9\times9}$, $\boldsymbol{B} \in \boldsymbol{R}^{9\times6}$	State transition matrix, input transition matrix
$\boldsymbol{H} \in \boldsymbol{R}^{6\times9}$	Measurement matrix
$\boldsymbol{P} \in \boldsymbol{R}^{9\times9}$	State estimate error covariance matrix
$\boldsymbol{W} \in \boldsymbol{R}^{9\times9}$	Disturbance estimate error covariance matrix
$\boldsymbol{C} \in \boldsymbol{R}^{9\times9}$	Estimate error cross-covariance matrix $\boldsymbol{C} = E\left[\left(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k\right)\left(\boldsymbol{w}_k - \hat{\boldsymbol{w}}_k\right)^T\right]$
$\boldsymbol{l} \in \boldsymbol{R}^9$	Disturbance estimating KF process error
$\boldsymbol{r} \in \boldsymbol{R}^9$	Disturbance estimating KF measurement noise
$\boldsymbol{Q}_l \in \boldsymbol{R}^{9\times9}$	Disturbance estimating KF process error covariance matrix
$\boldsymbol{R}_{\mathrm{m}} \in \boldsymbol{R}^{6\times6}$	State estimating KF measurement noise covariance matrix
$\boldsymbol{R}_r \in \boldsymbol{R}^{9\times9}$	Disturbance estimating KF measurement noise covariance matrix

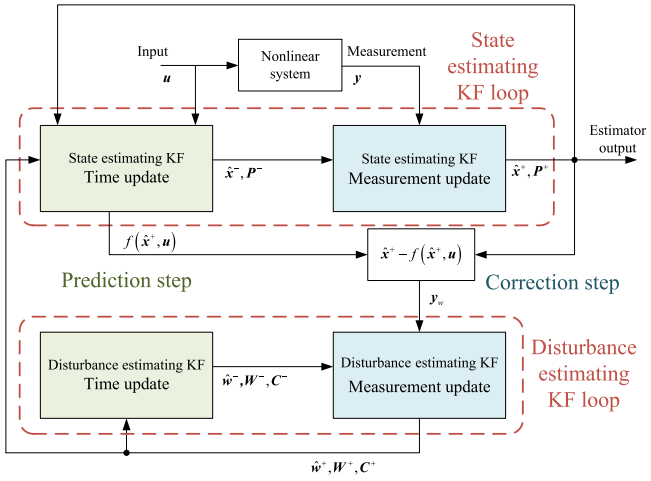


Fig. 2. DLKF overall algorithm structure diagram, from top to bottom are state estimating KF loop and disturbance estimating KF loop, from left to right are the overall prediction step and update step respectively.

robust state estimator, to compensate for this error. This section gives a brief introduction to the DLKF. For a detailed description and frequency analysis of the DLKF, see [24]. The related mathematical notations are shown in Table 1, and $\hat{\cdot}$ represents the estimated value, $+$ represents the posterior value, and $-$ represents the prior value. The relevant dimensions are given according to the model in Section 3.2.

For a nonlinear system, the discrete state space equation is:

$$\begin{aligned} \boldsymbol{x}_k &= f\left(\boldsymbol{x}_{k-1}, \boldsymbol{u}_{k-1}\right) + \boldsymbol{w}_{k-1} \\ \boldsymbol{y}_k &= h\left(\boldsymbol{x}_k\right) + \boldsymbol{v}_k \end{aligned} \tag{9}$$

Linearize Eq. (9) to matrix form, and the time interval is ΔT:

$$\begin{aligned} \boldsymbol{x}_k &= \boldsymbol{A}_{k-1}\boldsymbol{x}_{k-1} + \boldsymbol{B}_{k-1}\boldsymbol{u}_{k-1} + \boldsymbol{w}_{k-1} \\ \boldsymbol{y}_k &= \boldsymbol{H}_k\boldsymbol{x}_k + \boldsymbol{v}_k \end{aligned} \tag{10}$$

For the traditional EKF, $\boldsymbol{w}$ is treated as zero-mean, Gaussian, and non-correlated process error. Therefore, an error is introduced in the humanoid robots state estimation using simplified models. In addition to the original state estimating KF loop, The DLKF adds a disturbance estimating KF loop to estimate $\boldsymbol{w}$ in real time, which compensates for the estimation error caused by model errors and makes the state estimator more robust to model errors. The process model and measurement model of disturbance estimating KF loop are:

$$\begin{aligned} \boldsymbol{w}_k &= \boldsymbol{A}_{w,k-1}\boldsymbol{w}_{k-1} + \boldsymbol{l}_{k-1} \\ \boldsymbol{y}_{w,k} &= \boldsymbol{w}_k + \boldsymbol{r}_k \end{aligned} \tag{11}$$

where $\boldsymbol{A}_w$ is the identity matrix. This includes an assumption that the process error varies slowly and may have a non zero mean. For humanoid robots' CoM state estimation problem, there is a process error between the simplified model and the real dynamics of the robot, which has a non-zero mean and changes much slower than the filter period, so this assumption is reasonable. The measurement $\boldsymbol{y}_{w,k} = \hat{\boldsymbol{x}}_k^+ - f\left(\hat{\boldsymbol{x}}_{k-1}^+, \boldsymbol{u}_{k-1}\right)$ is calculated by the state estimating KF loop.

The calculation process of the DLKF can be summarized as the prediction step and update step in Table 2. The algorithm structure is shown in Fig. 2.

Table 2
The prediction step and update step of the DLKF.

Prediction step:
① State prediction step:
$\hat{\boldsymbol{x}}_k^- = f\left(\hat{\boldsymbol{x}}_{k-1}^+, \boldsymbol{u}_{k-1}\right) + \hat{\boldsymbol{w}}_{k-1}^+$
② Priori state estimate error covariance matrix calculation:
$\boldsymbol{P}_k^- = \boldsymbol{A}_{k-1}\boldsymbol{P}_{k-1}^+\boldsymbol{A}_{k-1}^T + \boldsymbol{A}_{k-1}\boldsymbol{C}_{k-1}^+ + \boldsymbol{C}_{k-1}^{+T}\boldsymbol{A}_{k-1}^T + \boldsymbol{W}_{k-1}^+$
③ Disturbance prediction step:
$\hat{\boldsymbol{w}}_k^- = \boldsymbol{A}_{w,k-1}\boldsymbol{w}_{k-1}^+$
④ Priori disturbance estimate error covariance matrix calculation:
$\boldsymbol{W}_k^- = \boldsymbol{A}_{w,k-1}\boldsymbol{W}_{k-1}^+\boldsymbol{A}_{w,k-1}^T + \boldsymbol{Q}_l$
⑤ Priori estimate error cross-covariance matrix calculation:
$\boldsymbol{C}_k^- = \boldsymbol{A}_{k-1}\boldsymbol{C}_{k-1}^+\boldsymbol{A}_{w,k-1}^T + \boldsymbol{W}_{k-1}^+\boldsymbol{A}_{w,k-1}^T$
Update step:
① State Kalman gain calculation:
$\boldsymbol{K}_k = \boldsymbol{P}_k^-\boldsymbol{H}_k^T\left(\boldsymbol{H}_k\boldsymbol{P}_k^-\boldsymbol{H}_k^T + \boldsymbol{R}_{\mathrm{m},k}\right)^{-1}$
② State estimate correction step:
$\hat{\boldsymbol{x}}_k^+ = \hat{\boldsymbol{x}}_k^- + \boldsymbol{K}_k\left(\boldsymbol{y}_k - h\left(\hat{\boldsymbol{x}}_k^-\right)\right)$
③ Update state estimate error covariance matrix:
$\boldsymbol{P}_k^+ = \left(\boldsymbol{I} - \boldsymbol{K}_k\boldsymbol{H}_k\right)\boldsymbol{P}_k^-$
④ Disturbance Kalman gain calculation:
$\boldsymbol{G}_k = \boldsymbol{C}_k^{-T}\boldsymbol{H}_k^T\boldsymbol{K}_k^T\left(\boldsymbol{K}_k\boldsymbol{H}_k\boldsymbol{P}_k^-\boldsymbol{H}_k^T\boldsymbol{K}_k^T + \boldsymbol{K}_k\boldsymbol{R}_k\boldsymbol{K}_k^T + \boldsymbol{R}_r\right)^{-1}$
⑤ Disturbance estimate correction step:
$\hat{\boldsymbol{w}}_k^+ = \hat{\boldsymbol{w}}_k^- + \boldsymbol{G}_k\left(\boldsymbol{y}_{w,k} - \hat{\boldsymbol{w}}_k^-\right)$
⑥ Update disturbance estimate error covariance matrix:
$\boldsymbol{W}_k^+ = \boldsymbol{W}_k^- - \boldsymbol{G}_k\boldsymbol{K}_k\boldsymbol{H}_k\boldsymbol{C}_k^- - \boldsymbol{C}_k^{-T}\boldsymbol{H}_k^T\boldsymbol{K}_k^T\boldsymbol{G}_k^T + \boldsymbol{G}_k\boldsymbol{P}_k^-\boldsymbol{H}_k^T\boldsymbol{K}_k^T\boldsymbol{G}_k^T + \boldsymbol{G}_k\boldsymbol{R}_r\boldsymbol{G}_k^T$
⑦ Update estimate error cross-covariance matrix:
$\boldsymbol{C}_k^+ = \left(\boldsymbol{I} - \boldsymbol{K}_k\boldsymbol{H}_k\right)\boldsymbol{C}_k^- - \left(\boldsymbol{I} - \boldsymbol{K}_k\boldsymbol{H}_k\right)\boldsymbol{P}_k^-\boldsymbol{H}_k^T\boldsymbol{K}_k^T\boldsymbol{G}_k^T + \boldsymbol{K}_k\boldsymbol{R}_k\boldsymbol{K}_k^T\boldsymbol{G}_k^T$

3. Nonlinear state estimation framework

The goal of our nonlinear state estimation framework is to estimate the CoM state. It consists of the estimation of floating base kinematics, and the estimation of CoM state. Subsequently, with the estimation of the floating base kinematics, the estimation of the CoM state can be realized.

3.1. Floating base kinematics estimation based on the IEKF

For a mobile robot, an accurate and fast-converging floating base kinematics estimator is very helpful to humanoid robot state estimation. The IEKF(Invariant extended Kalman filter) is implemented in this work. The IEKF utilizes the property that group affine of the certain matrix Lie groups, realizing the linear error passing with nonlinear process model, which can both ensure accuracy and fast convergence [14]. This section summarizes the floating base kinematics estimator based on the IEKF. Further details have been discussed in [16,17]. The involved mathematical notations are shown in Table 3, in which I stands for IEKF.

The contact foot is first detected through the FSRs force measurements,

$$f_{N,*} > f_{th} \tag{12}$$

where, $f_{N,*}$ is the vertical contact force of foot *, f_{th} is the threshold. When Eq. (12) is true, we believe that foot * is the contact foot.

The state of the estimator consists of 2 parts, one is a matrix Lie group that includes $\boldsymbol{R}_{\mathrm{WB}}$, ${}_{\mathrm{W}}\boldsymbol{v}_{\mathrm{WB}}$, ${}_{\mathrm{W}}\boldsymbol{p}_{\mathrm{WB}}$, ${}_{\mathrm{W}}\boldsymbol{d}_{\mathrm{WC}}$, as shown in Eq. (13) and Fig. 3, and the subscript is simplified for the sake of readability.

$$\boldsymbol{X} \triangleq \begin{bmatrix} \boldsymbol{R} & \boldsymbol{v} & \boldsymbol{p} & \boldsymbol{d} \\ \boldsymbol{0}_{1,3} & 1 & 0 & 0 \\ \boldsymbol{0}_{1,3} & 0 & 1 & 0 \\ \boldsymbol{0}_{1,3} & 0 & 0 & 1 \end{bmatrix} \in \mathrm{SE}_3(3). \tag{13}$$

Table 3
Mathematical notations for the IEKF.

Mathematical notation	Illustration
$\boldsymbol{R}_{\mathrm{WB}} \in \boldsymbol{R}^{3\times3}$	Floating base rotation matrix in the world frame
${}_{\mathrm{W}}\boldsymbol{v}_{\mathrm{WB}} \in \boldsymbol{R}^3$	Floating base velocity in the world frame
${}_{\mathrm{W}}\boldsymbol{p}_{\mathrm{WB}} \in \boldsymbol{R}^3$	Floating base position in the world frame
${}_{\mathrm{W}}\boldsymbol{d}_{\mathrm{WC}} \in \boldsymbol{R}^3$	Contact frame origin position in the world frame
$\boldsymbol{b}^{\mathrm{g}} \in \boldsymbol{R}^3$	IMU gyroscope bias
$\boldsymbol{b}^{\mathrm{a}} \in \boldsymbol{R}^3$	IMU accelerometer bias
$\boldsymbol{\omega} \in \boldsymbol{R}^3$	Angular velocity measurement from IMU
$\boldsymbol{a} \in \boldsymbol{R}^3$	Acceleration measurement from IMU
$\boldsymbol{g} \in \boldsymbol{R}^3$	Acceleration of gravity
$\boldsymbol{w}^{\mathrm{g}} \in \boldsymbol{R}^3$	Zero mean Gaussian noise from IMU gyroscope
$\boldsymbol{w}^{\mathrm{a}} \in \boldsymbol{R}^3$	Zero mean Gaussian noise from IMU accelerometer
$\boldsymbol{w}^{\mathrm{bg}} \in \boldsymbol{R}^3$	Zero mean Gaussian noise from IMU gyroscope bias
$\boldsymbol{w}^{\mathrm{ba}} \in \boldsymbol{R}^3$	Zero mean Gaussian noise from IMU accelerometer bias
$\boldsymbol{w}^{\mathrm{v}} \in \boldsymbol{R}^3$	Slip velocity of the contact frame
$\boldsymbol{\alpha} \in \boldsymbol{R}^N$	Joint angles, with N joints
${}_{\mathrm{B}}\boldsymbol{R}_{\mathrm{BC}} \in \boldsymbol{R}^{3\times3}$	Orientation of the contact frame with respect to floating base
$\boldsymbol{w}_{\mathrm{I}}$	$vec\left(\boldsymbol{w}^{\mathrm{g}}, \boldsymbol{w}^{\mathrm{a}}, \boldsymbol{0}_{3,1}, {}_{\mathrm{B}}\boldsymbol{R}_{\mathrm{BC}}\left(\boldsymbol{\alpha}_k\right)\boldsymbol{w}^{\mathrm{v}}, \boldsymbol{w}^{\mathrm{bg}}, \boldsymbol{w}^{\mathrm{ba}}\right)$
$\boldsymbol{J}_{\mathrm{v}}(\cdot)$	Geometric Jacobian corresponding to linear velocity
$\boldsymbol{w}_k^{\alpha} \in \boldsymbol{R}^N$	Zero mean Gaussian noise from encoders
$\boldsymbol{p}_{\mathrm{BC}} \in \boldsymbol{R}^3$	Position of the contact frame with respect to floating base

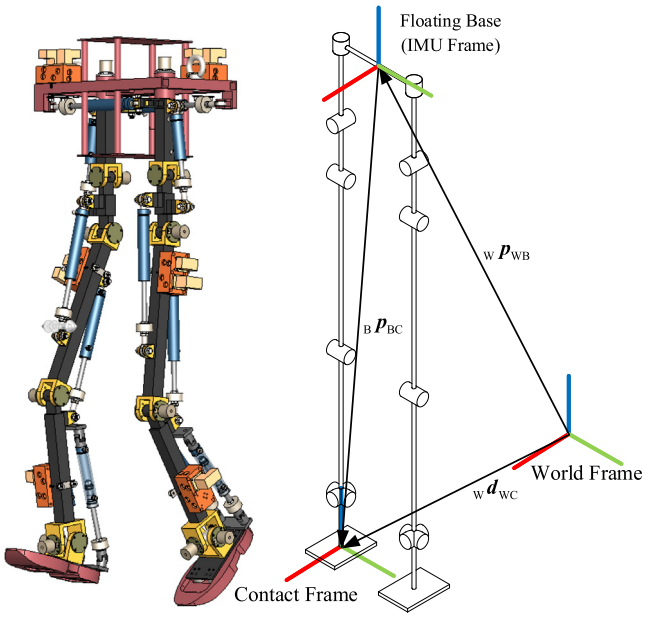


Fig. 3. The three-dimensional structure of the Defensor hydraulic biped robot (left). The simplified structure of the Defensor hydraulic biped robot (right). The robot has 12 actuated degrees of freedom and 6 degrees of freedom on a single leg, which are 3 degrees of freedom for the hip, 1 for the knee, and 2 for the ankle.

The other part that cannot be merged into matrix Lie group is formed into a vector,

$$\boldsymbol{\theta} \triangleq \begin{bmatrix} \boldsymbol{b}^{\mathrm{g}} \\ \boldsymbol{b}^{\mathrm{a}} \end{bmatrix} \in \boldsymbol{R}^6. \tag{14}$$

The prediction step of the estimator is as follows.

State prediction step:

$$\begin{aligned}\hat{\boldsymbol{R}}_k^- &= \hat{\boldsymbol{R}}_{k-1}^+ \exp\left(\left(\boldsymbol{\omega}_k - \hat{\boldsymbol{b}}_{k-1}^{g+}\right)\Delta t\right)\\ \hat{\boldsymbol{v}}_k^- &= \hat{\boldsymbol{v}}_{k-1}^+ + \hat{\boldsymbol{R}}_{k-1}^+\left(\boldsymbol{a}_k - \hat{\boldsymbol{b}}_{k-1}^{a+}\right)\Delta t + \boldsymbol{g}\Delta t\\ \hat{\boldsymbol{p}}_k^- &= \hat{\boldsymbol{p}}_{k-1}^+ + \hat{\boldsymbol{v}}_{k-1}^+\Delta t + \tfrac{1}{2}\hat{\boldsymbol{R}}_{k-1}^+\left(\boldsymbol{a}_k - \hat{\boldsymbol{b}}_{k-1}^{a+}\right)\Delta t^2 + \tfrac{1}{2}\boldsymbol{g}\Delta t^2\\ \hat{\boldsymbol{d}}_k^- &= \hat{\boldsymbol{d}}_{k-1}^+, \quad \hat{\boldsymbol{b}}_k^{g-} = \hat{\boldsymbol{b}}_{k-1}^{g+}, \quad \hat{\boldsymbol{b}}_k^{a-} = \hat{\boldsymbol{b}}_{k-1}^{a+}\end{aligned} \tag{15}$$

Priori state estimate error covariance matrix calculation:

$$\begin{aligned}\boldsymbol{P}_{\mathrm{I},k}^- = &\exp\left(\boldsymbol{A}_{\mathrm{I},k-1}\Delta t\right)\boldsymbol{P}_{\mathrm{I},k-1}^+\exp\left(\boldsymbol{A}_{\mathrm{I},k-1}\Delta t\right)^\top\\ &+\exp\left(\boldsymbol{A}_{\mathrm{I},k-1}\Delta t\right)\boldsymbol{Q}_{k-1}\exp\left(\boldsymbol{A}_{\mathrm{I},k-1}\Delta t\right)^\top\Delta t.\end{aligned} \tag{16}$$

where $\boldsymbol{A}_{\mathrm{I},k} = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\hat{\boldsymbol{R}}_k^+ & \mathbf{0}\\ (\boldsymbol{g})_\times & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\left(\hat{\boldsymbol{v}}_k^+\right)_\times\hat{\boldsymbol{R}}_k^+ & -\hat{\boldsymbol{R}}_k^+\\ \mathbf{0} & \mathbf{I} & \mathbf{0} & \mathbf{0} & -\left(\hat{\boldsymbol{p}}_k^+\right)_\times\hat{\boldsymbol{R}}_k^+ & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & -\left(\hat{\boldsymbol{d}}_k^+\right)_\times\hat{\boldsymbol{R}}_k^+ & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0}\\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0}\end{bmatrix}$, $\boldsymbol{Q}_k = \begin{bmatrix}\mathrm{Ad}_{\hat{\boldsymbol{X}}_k^+} & \mathbf{0}_{12,6}\\ \mathbf{0}_{6,12} & \mathbf{I}_6\end{bmatrix} Cov\left(\boldsymbol{w}_{\mathrm{I},k}\right)\begin{bmatrix}\mathrm{Ad}_{\hat{\boldsymbol{X}}_k^+} & \mathbf{0}_{12,6}\\ \mathbf{0}_{6,12} & \mathbf{I}_6\end{bmatrix}^\top$.

The update step is as follows.

State Kalman gain calculation:

$$\boldsymbol{K}_{\mathrm{I},k} = \boldsymbol{P}_{\mathrm{I},k}^-\boldsymbol{H}_{\mathrm{I},k}^\top\left(\boldsymbol{H}_{\mathrm{I},k}\boldsymbol{P}_{\mathrm{I},k}^-\boldsymbol{H}_{\mathrm{I},k}^\top + \boldsymbol{N}_k\right)^{-1}. \tag{17}$$

where $\boldsymbol{H}_{\mathrm{I},k} = \begin{bmatrix}\mathbf{0} & \mathbf{0} & -\mathbf{I} & \mathbf{I} & \mathbf{0} & \mathbf{0}\end{bmatrix}$, $\boldsymbol{N}_k = \hat{\boldsymbol{R}}_k^-\mathbf{J}_\mathrm{v}\left(\boldsymbol{\alpha}_k\right)Cov\left(\boldsymbol{w}_k^\alpha\right)\mathbf{J}_\mathrm{v}^\top\left(\boldsymbol{\alpha}_k\right)\hat{\boldsymbol{R}}_k^{-\top}$.

State estimate correction:

$$\left(\hat{\boldsymbol{X}}_k^+, \boldsymbol{\theta}_t^+\right) = \left(\exp\left(\boldsymbol{K}_k^\xi\boldsymbol{I}\left(\hat{\boldsymbol{X}}_k^-\boldsymbol{Y}_k\right)\right)\hat{\boldsymbol{X}}_k^-, \hat{\boldsymbol{\theta}}_k^- + \boldsymbol{K}_k^\zeta\boldsymbol{I}\left(\hat{\boldsymbol{X}}_k^-\boldsymbol{Y}_k\right)\right). \tag{18}$$

where $\boldsymbol{K}_k^\xi, \boldsymbol{K}_k^\zeta$ are the corresponded row in $\boldsymbol{K}_{\mathrm{I},k}$. $\boldsymbol{Y}_k^\top = \begin{bmatrix}{}_\mathrm{B}\boldsymbol{p}_\mathrm{BC}\left(\boldsymbol{\alpha}_k\right) & 0 & 1 & -1\end{bmatrix}$.

Update state estimate error covariance matrix:

$$\boldsymbol{P}_k^+ = \left(\mathbf{I} - \boldsymbol{K}_{\mathrm{I},k}\boldsymbol{H}_{\mathrm{I},k}\right)\boldsymbol{P}_{\mathrm{I},k}^-\left(\mathbf{I} - \boldsymbol{K}_{\mathrm{I},k}\boldsymbol{H}_{\mathrm{I},k}\right)^\top + \boldsymbol{K}_{\mathrm{I},k}\boldsymbol{N}_k\boldsymbol{K}_{\mathrm{I},k}^\top. \tag{19}$$

It is notable that, due to the limitation of the sensors, $\boldsymbol{p}$ and the yaw part of $\boldsymbol{R}$ cannot be observed. In practice, they will diverge as error accumulates.

3.2. CoM estimation based on nonlinear dynamics and DLKF

This section will explain in detail the CoM state estimator based on nonlinear dynamics and DLKF. The mathematical notations correspond to those in Sections 2 and 3.1.

The system state includes CoM position, CoM velocity and external forces. The state vector is defined as:

$$\boldsymbol{x} = \begin{bmatrix}c^x & c^y & c^z & \dot{c}^x & \dot{c}^y & \dot{c}^z & f_e^x & f_e^y & f_e^z\end{bmatrix}^\top. \tag{20}$$

The system input includes the ZMP position, the vertical contact force with the ground and the rate of angular momentum, and the input vector is defined as:

$$\boldsymbol{u} = \begin{bmatrix}z^x & z^y & z^z & f_N & \dot{L}^x & \dot{L}^y\end{bmatrix}^\top \tag{21}$$

where f_N can be directly measured by FSRs, and the position of the ZMP can be calculated according to Eq. (3). f_i^z is the vertical contact force measured by a single FSR, $\boldsymbol{p}_i = {}_\mathrm{W}\boldsymbol{p}_\mathrm{WB} + \boldsymbol{R}_\mathrm{WB}{}_\mathrm{B}\boldsymbol{p}_\mathrm{Bi}\left(\boldsymbol{\alpha}\right)$, ${}_\mathrm{B}\boldsymbol{p}_\mathrm{Bi}$ is the position of the ith FSR in the floating base. ${}_\mathrm{W}\boldsymbol{p}_\mathrm{WB}$ and $\boldsymbol{R}_\mathrm{WB}$ are obtained by floating base kinematics estimator.

We assume that the inertia of the robot rotating around the floating base is constant, represented by $\boldsymbol{I}_\mathrm{b}$, then:

$$\dot{\boldsymbol{L}} = \boldsymbol{I}_\mathrm{b}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega}\times\boldsymbol{I}_\mathrm{b}\boldsymbol{\omega}. \tag{22}$$

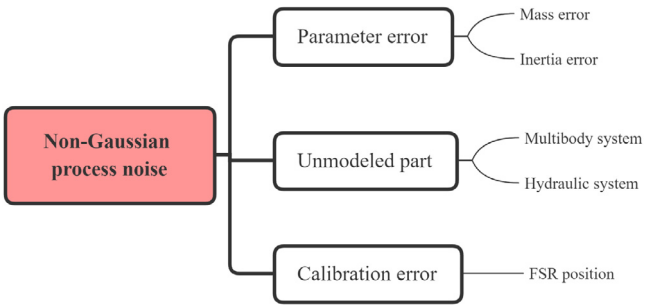


Fig. 4. Sources of non-Gaussian noise.

The measurement of the system includes CoM position and CoM acceleration, which is defined as:

$$\boldsymbol{y} = \begin{bmatrix}c^x & c^y & c^z & \ddot{c}^x & \ddot{c}^y & \ddot{c}^z\end{bmatrix}^\top \tag{23}$$

where, $\boldsymbol{c}$ is jointly calculated by floating base kinematics estimator, forward kinematics and CAD(Computer Aided Design) model. Moreover, assuming that CoM and IMU are located on the same rigid body (For humanoid robots, IMU is often located at the torso and the CoM can be approximately considered to be on the torso, thus this assumption is reasonable), $\ddot{\boldsymbol{c}} = \boldsymbol{a} + \boldsymbol{\omega}\times(\boldsymbol{\omega}\times\boldsymbol{r}_\boldsymbol{c}) + \dot{\boldsymbol{\omega}}\times\boldsymbol{r}_\boldsymbol{c}$ [20], where $\boldsymbol{r}_\boldsymbol{c}$ is the vector from IMU to CoM computed using forward kinematics.

From Eqs. (6) to (8), the nonlinear dynamic equation of the state $\boldsymbol{x}$ is:

$$\dot{\boldsymbol{x}} = g\left(\boldsymbol{x}, \boldsymbol{u}\right) + \boldsymbol{w}_x \tag{24}$$

$$g\left(\boldsymbol{x}, \boldsymbol{u}\right) = \begin{bmatrix}\dot{c}^x\\ \dot{c}^y\\ \dot{c}^z\\ \frac{c^x - z^x}{m(c^z - z^z)}f_N - \frac{\dot{L}^y}{m(c^z - z^z)} + \frac{1}{m}f_e^x\\ \frac{c^y - z^y}{m(c^z - z^z)}f_N + \frac{\dot{L}^x}{m(c^z - z^z)} + \frac{1}{m}f_e^y\\ \frac{1}{m}\left(f_N + f_e^z\right) - g^z\\ 0\\ 0\\ 0\end{bmatrix} + \boldsymbol{w}_x \tag{25}$$

where $\boldsymbol{w}_x$ is the noise term. First, discretize Eq. (24) to obtain the state transition function used for the prior value calculation:

$$f\left(\boldsymbol{x}, \boldsymbol{u}\right) = \boldsymbol{I}^9 + g\left(\boldsymbol{x}, \boldsymbol{u}\right)\Delta t. \tag{26}$$

Then the first-order Taylor approximation is performed on Eq. (24) to obtain the state transition matrix for DLKF:

$$\boldsymbol{A} = \boldsymbol{I}^{9\times9} + \frac{\partial g}{\partial \boldsymbol{x}}\Delta t. \tag{27}$$

where,

$$\frac{\partial g}{\partial \boldsymbol{x}} = \begin{bmatrix}\mathbf{0}^{3\times3} & \boldsymbol{I}^{3\times3} & \mathbf{0}^{3\times3}\\ \boldsymbol{G}_1 & \mathbf{0}^{3\times3} & \boldsymbol{G}_2\\ \mathbf{0}^{3\times3} & \mathbf{0}^{3\times3} & \mathbf{0}^{3\times3}\end{bmatrix},$$

$$\boldsymbol{G}_1 = \begin{bmatrix}\frac{f_N}{m(c^z - z^z)} & 0 & -\frac{(c^x - z^x)f_N - \dot{L}^y}{m(c^z - z^z)^2}\\ 0 & \frac{f_N}{m(c^z - z^z)} & -\frac{(c^y - z^y)f_N + \dot{L}^x}{m(c^z - z^z)^2}\\ 0 & 0 & 0\end{bmatrix},$$

$$\boldsymbol{G}_2 = \begin{bmatrix}\frac{1}{m} & 0 & 0\\ 0 & \frac{1}{m} & 0\\ 0 & 0 & \frac{1}{m}\end{bmatrix}.$$

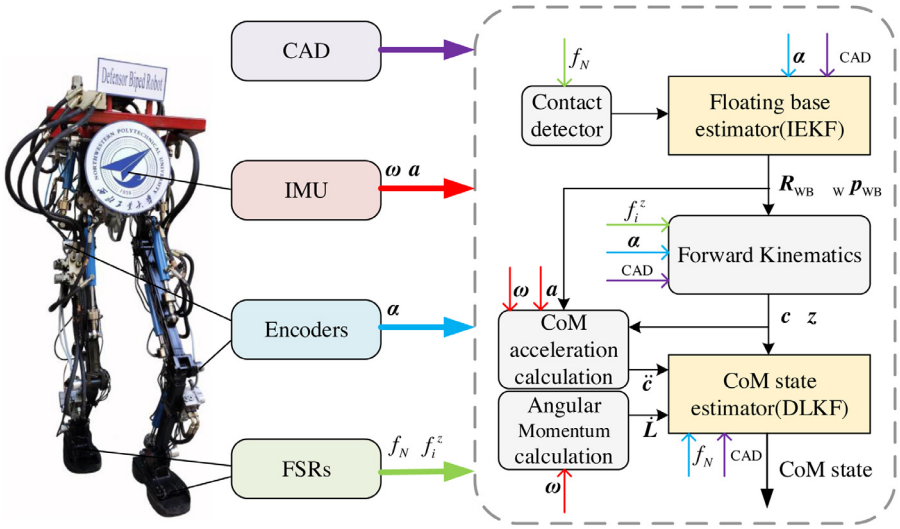


Fig. 5. The overall frame diagram of the humanoid robot nonlinear state estimator, the left side is the position of the relevant sensors on the Defensor hydraulic biped robot, the right side is the algorithm flow, the yellow part is the filter, and the gray part is the auxiliary calculation.

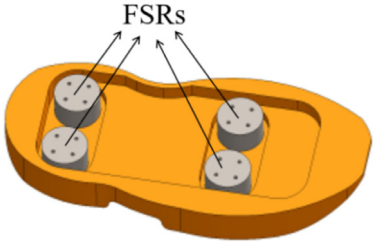


Fig. 6. 4 FSRs mounted on foot's sole.

In the same way, a nonlinear measurement model can be obtained:

$$\boldsymbol{y} = h(\boldsymbol{x}, \boldsymbol{u}) + \boldsymbol{v}, \tag{28}$$

$$h(\boldsymbol{x}, \boldsymbol{u}) = \begin{bmatrix} c^x \\ c^y \\ c^z \\ \frac{c^x - z^x}{m(c^z - z^z)} f_N - \frac{\dot{L}^y}{m(c^z - z^z)} + \frac{1}{m} f_e^x \\ \frac{c^y - z^y}{m(c^z - z^z)} f_N + \frac{\dot{L}^x}{m(c^z - z^z)} + \frac{1}{m} f_e^y \\ \frac{1}{m}\left(f_N + f_e^z\right) - g^z \end{bmatrix}. \tag{29}$$

Perform first-order Taylor approximation to Eq. (28) to obtain the measurement matrix for DLKF:

$$\boldsymbol{H} = \begin{bmatrix} \boldsymbol{I}^{3\times3} & \boldsymbol{0}^{3\times3} & \boldsymbol{0}^{3\times3} \\ \boldsymbol{H}_1 & \boldsymbol{0}^{3\times3} & \boldsymbol{H}_2 \end{bmatrix}. \tag{30}$$

where,

$$\boldsymbol{H}_1 = \begin{bmatrix} \frac{f_N}{m(c^z - z^z)} & 0 & -\frac{(c^x - z^x) f_N - \dot{L}^y}{m(c^z - z^z)^2} \\ 0 & \frac{f_N}{m(c^z - z^z)} & -\frac{(c^y - z^y) f_N + \dot{L}^x}{m(c^z - z^z)^2} \\ 0 & 0 & 0 \end{bmatrix},$$

$$\boldsymbol{H}_2 = \begin{bmatrix} \frac{1}{m} & 0 & 0 \\ 0 & \frac{1}{m} & 0 \\ 0 & 0 & \frac{1}{m} \end{bmatrix}.$$

This process model uses the full nonlinear CoM dynamics of humanoid robots. However, it also brings non-zero-mean, non-Gaussian, correlated error that cannot be compensated by traditional EKF as shown in Fig. 4. There are three sources of the non-Gaussian noise as follows. Initially, there are errors between model parameters and real parameters, including mass m and inertia $\boldsymbol{I}_b$. It is not realistic to simply regard inertia as a constant during the movement of a multi-rigid body system. Moreover, this process model has many unmodeled parts such as multi-rigid body system and drive system (For defensor humanoid robot, it is the hydraulic system). Furthermore, there are calibration errors between the real prototype and the CAD model. The ZMP is computed by Eq. (3). When there is an error in the position of the FSR, the non-Gaussian noise will also be introduced in the prediction step.

As a result, it is necessary to use the DLKF to compensate for the non-Gaussian noise in the process model. Substituting Eqs. (25) to (30) into the DLKF in Section 2.2, the DLKF can be directly used with the nonlinear model to compensate for process error. The entire nonlinear state estimation framework is shown in Fig. 5. Moreover, in order to further explain the algorithm flow, the nonlinear state estimation framework for humanoid robots proposed in this paper is summarized in the Appendix using pseudo-code.

4. Simulation

This section will verify and compare the algorithms of the nonlinear state estimation framework proposed in this paper. In the comparison process, the true state of the robot needs to be obtained to measure the estimation error of the algorithm. Due to many uncontrollable factors in the real environment, the accurate robot state is difficult to be measured. Therefore, we verify the algorithm in a virtual environment. Simscape Multibody [31] is used as the virtual environment, and the real robot state can be easily read and calculated from internal variables. This section first describes the framework and simulation conditions of the algorithm verification, then compares the accuracy of the CoM state and DCM estimation between different algorithms. Finally, the IEKF is used for the estimation of the floating base, and the whole nonlinear state estimation framework is simulated.

4.1. Validation framework and conditions

In this paper, Defensor hydraulic biped robot developed by our team is used for algorithm verification. The three-dimensional structure and simplified structure diagram are shown in Fig. 3,

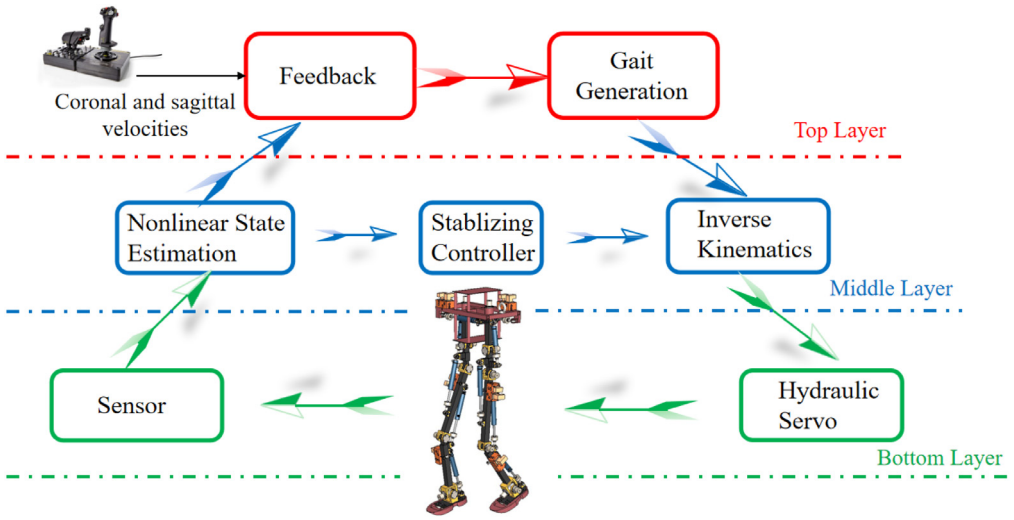


Fig. 7. Hierarchical control architecture. The left side is the feedback path. The right side is the control path.

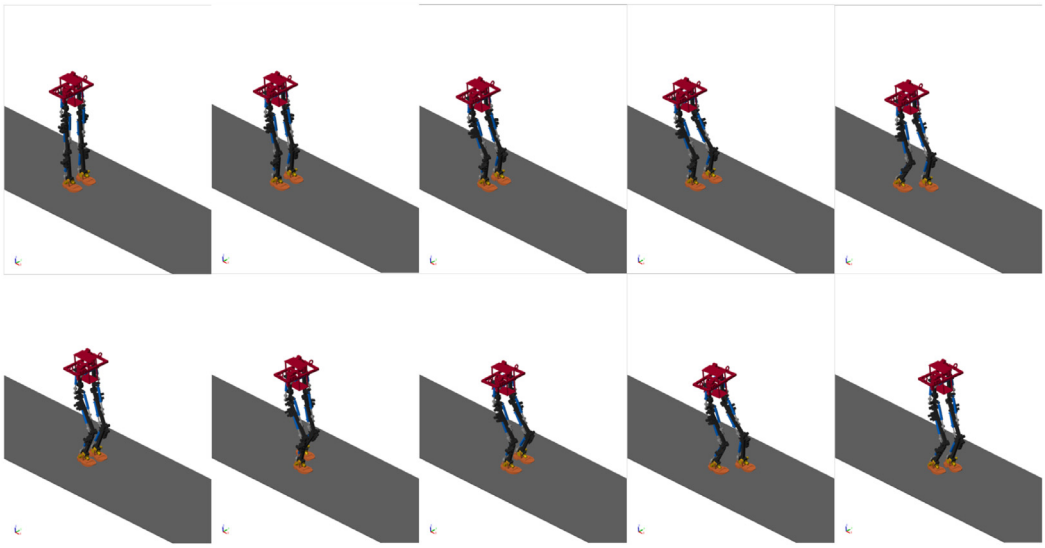


Fig. 8. Defensor hydraulic biped robot walks in the virtual environment.

and the physical prototype is shown on the left part of Fig. 5. The waist of the robot is equipped with an IMU whose position coincides with the floating base. Each joint is equipped with an absolute angular displacement encoder. There are a total of 12 encoders in the whole body. Each foot has 4 FSRs, as shown in Fig. 6.

The walking control of the robot adopts the hierarchical control method [32,33], and the control architecture is shown in Fig. 7. The input instruction is the reference speed of the robot in the coronal and sagittal directions. The top layer plans the robot gait. This article uses the model predictive control (MPC)-based gait planning method in [3], and the middle layer computes the inverse kinematics, stability control and state estimation of the robot. Inverse kinematics solution and stability control adopt the numerical method based on the centroid Jacobian matrix in [34]. The state estimator is the nonlinear state estimator proposed in this paper. The bottom layer is the virtual prototype model of the robot in the virtual environment, and the hydraulic part is simulated by Simscape Fluids [35].

The algorithm verification is carried out on a personal laptop, and the relevant software and hardware are shown in Table 4.

Table 4
Software and hardware.

Item	Illustration
Operating system	Windows 10 64 Bit Home Chinese Version
CAD software	SolidWorks 2020 64 bit
Simulation software	Matlab/Simulink R2020b 64 bit
CPU	i7-7700HQ
RAM	16 GB
Hard disk	1Tb SSD

The simulation parameters are shown in Table 5, where h_c is the planned CoM height.

The robot walks for a total of 20 s. The reference speed of the sagittal direction is 0 m/s from 0 to 1 s, 0.1 m/s for 1 to 4 s, and then 0.2 m/s. The reference speed of the coronal direction is always 0 m/s. The visual movement of the robot in the virtual environment is shown in Fig. 8.

The algorithm verification flowchart is shown in Fig. 9. The sensor data with virtual noise obtained from the virtual environment enter the estimator. Considering in the real environment, due to material and calibration errors, the quality parameters

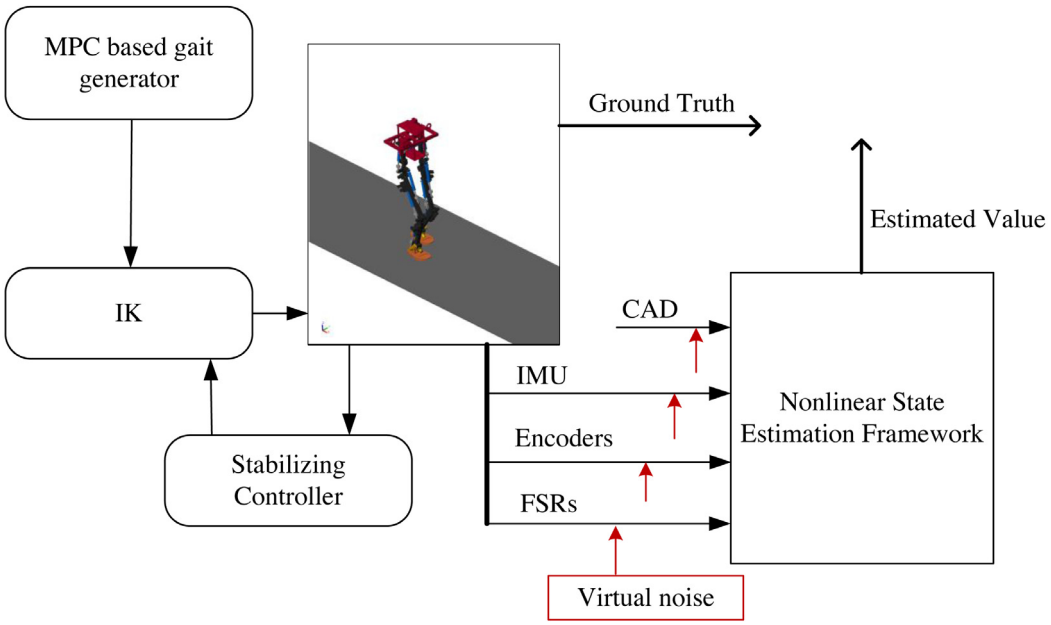


Fig. 9. Algorithm verification flowchart. The red part is the added virtual noise.

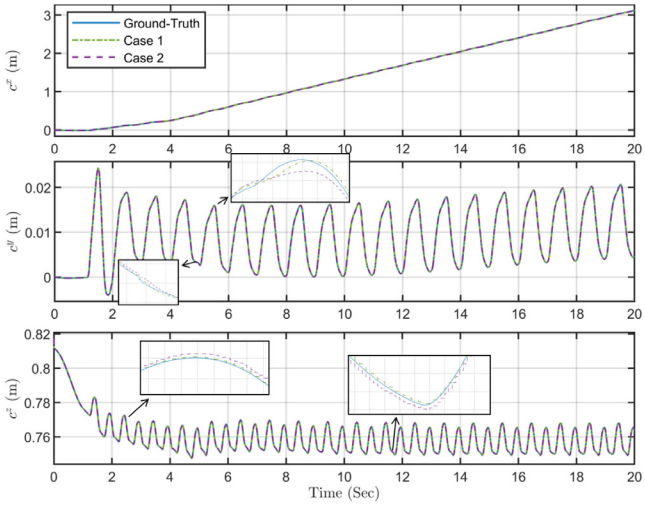


Fig. 10. CoM position estimation in the x, y and z directions.

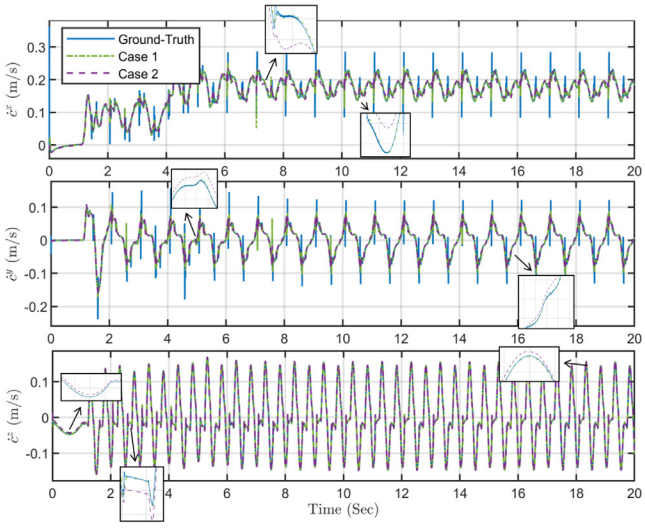


Fig. 11. CoM velocity estimation in the x, y and z directions.

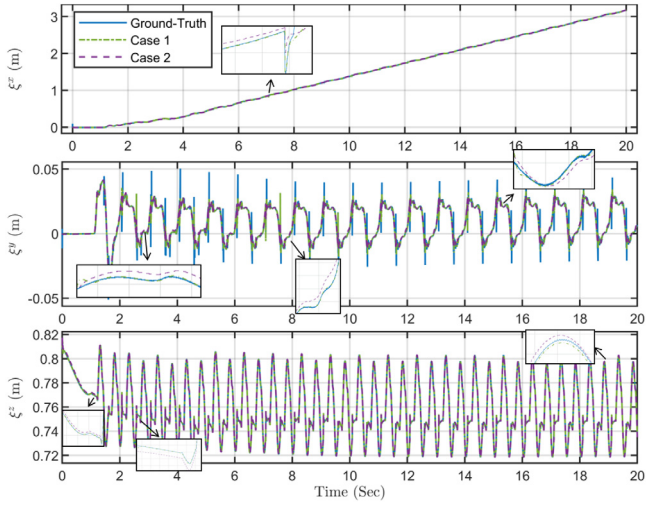


Fig. 12. DCM position estimation in the x, y and z directions.

of real robots are often different from the quality parameters obtained from the CAD model. In order to reflect the model errors, we use the fine-tuned relevant quality parameters obtained from the CAD model.

4.2. Comparison

In this scenario, we compare our CoM state estimation method with previous methods. The following two situations will be mainly compared:

Case 1: Nonlinear dynamics + DLKF

Case 2: Nonlinear dynamics + EKF

For fair performance evaluation, the same measurement noise covariance is used for both cases. The noise covariances are shown in Table 6. In order to eliminate the influence of the floating base kinematics estimator, the true values of $\boldsymbol{R}_{\mathrm{WB}}$ and ${}_{\mathrm{W}}\boldsymbol{p}_{\mathrm{WB}}$ are directly used to calculate the positions of CoM and ZMP. Fig. 10 shows the estimation of the CoM position in the x, y and z directions, and Fig. 11 shows the estimation of the CoM velocity

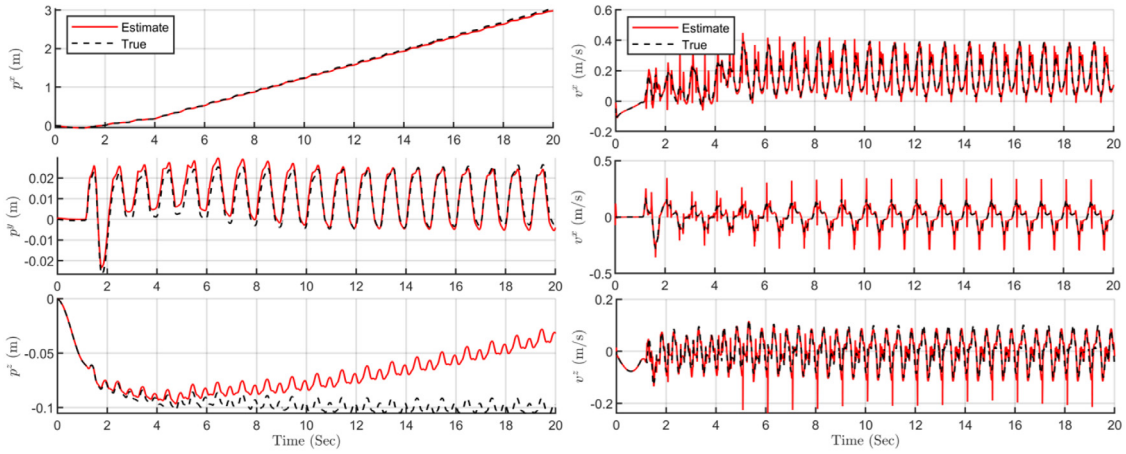


Fig. 13. The position (left) and velocity (right) of the floating base in the world coordinate system estimated by IEKF.

Table 5
Simulation parameters.

Physical quantity	Value
m (kg)	114.8
I_b^{xx} (kg m^2)	0.5
I_b^{yy} (kg m^2)	0.3
I_b^{zz} (kg m^2)	0.5
h_c (m)	0.76

Table 6
Filter parameters.

Case 1						Case 2				
$\boldsymbol{Q}_l$			$\boldsymbol{R}_r$			$\boldsymbol{Q}$			$\boldsymbol{R}_m$	
$\boldsymbol{c}$	$\dot{\boldsymbol{c}}$	$\boldsymbol{f}_e$	$\boldsymbol{c}$	$\dot{\boldsymbol{c}}$	$\boldsymbol{f}_e$	$\boldsymbol{c}$	$\dot{\boldsymbol{c}}$	$\boldsymbol{f}_e$	$\boldsymbol{c}$	$\ddot{\boldsymbol{c}}$
$1e^{-20}$	$1e^{-8}$	$1e^{-10}$	$1e^{-5}$	$1e^{-10}$	$1e^{-5}$	$1e^{-12}$	$1e^{-6}$	$1e^{-4}$	$1e^{-8}$	$4e^{-2}$

in the x, y and z directions. Moreover, in addition to the CoM state feedback, many controllers also use DCM as the feedback to control the walking stability of the humanoid robot [36]. We also compare the performance of the DCM estimation in the two cases in Fig. 12. The DCM is given by:

$$\boldsymbol{\xi} = \mathbf{c} + \frac{1}{\omega_0}\dot{\mathbf{c}} \tag{31}$$

where $\omega_0 = \sqrt{\frac{g}{h_c}}$.

The results show that because DLKF better compensates for the non-zero-mean, non-Gaussian and correlated process error caused by the parameter error, unmodeled part and calibration error, in the x, y and z directions, the CoM position, CoM velocity and DCM position of Case 1 are closer to the ground truth. Case 1 shows better state estimation performance.

In order to further intuitively reflect the filter performance, we calculate the root mean square error (RMSE). The linear model is also considered in the following two cases:

Case 3: Linear dynamics + DLKF

Case 4: Linear dynamics + KF

The Eqs. (12) and (19) in [19] are used as linear model. The results are shown in Table 7. It should be noted that in the z axis no linear estimate is available.

From Case 4 to Case 1, the performance gradually gets better. The performance improvement of the estimator comes from two parts, nonlinear model and DLKF. Among them, the nonlinear model has a greater effect on accuracy. The Nonlinear dynamics + DLKF (Case1) proposed in this paper achieves the best estimation

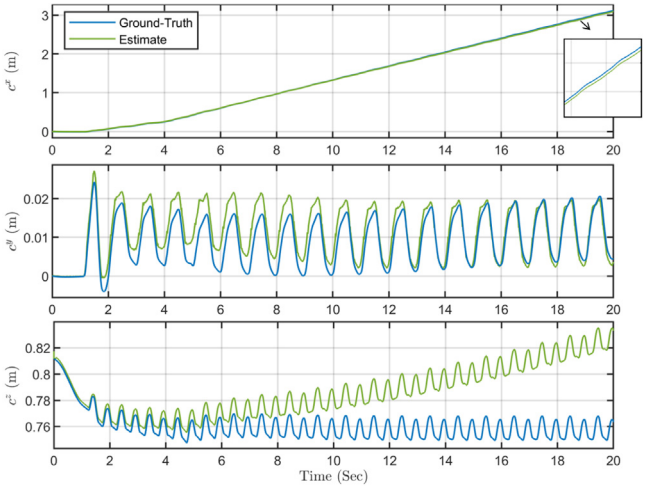


Fig. 14. CoM position estimation of the whole nonlinear state estimation framework in the x, y and z directions.

performance. Compare with Case 3. In the x direction, the CoM position RMSE is reduced by 96%, the CoM velocity RMSE is reduced by 96%, and the DCM position RMSE is reduced by 96%. In the y direction, the CoM position RMSE is reduced by 93%, the CoM velocity RMSE is reduced by 96%, and the DCM position RMSE is reduced by 96%. Compare with Case 2. In the x direction, the CoM position RMSE is reduced by 4%, the CoM velocity RMSE is reduced by 3%, and the DCM position RMSE is reduced by 4%. In the y direction, the CoM position RMSE is reduced by 33%, the CoM velocity RMSE is reduced by 38%, and the DCM position RMSE is reduced by 34%. In the z direction, the CoM position RMSE is reduced by 2%, the CoM velocity RMSE is reduced by 28%, and the DCM position RMSE is reduced by 29%.

4.3. Whole framework simulation

In this scenario, we verify the whole nonlinear estimation framework. The floating base kinematics are estimated by the IEKF as shown in Fig. 13, which will be used to calculate the measurement positions of the ZMP and the CoM, and the acceleration of the CoM. It can be seen that due to the unobservability of the floating base position, as the integration error accumulates, the estimation error will diverge.

The result of the CoM state estimation is shown in Fig. 14. In the initial stage, the framework can estimate a relatively accurate

Table 7
RMSE for CoM position, CoM velocity and DCM position.

	c^x (m)	c^y (m)	c^z (m)	$\dot{c}^x$ (m/s)	$\dot{c}^y$ (m/s)	$\dot{c}^z$ (m/s)	ξ^x (m)	ξ^y (m)	ξ^z (m)
Case 1	$5.3e^{-5}$	$4.1e^{-5}$	$4.9e^{-5}$	$8.3e^{-3}$	$8.7e^{-3}$	$1.8e^{-3}$	$2.3e^{-3}$	$2.5e^{-3}$	$5.0e^{-4}$
Case 2	$5.5e^{-5}$	$6.1e^{-5}$	$5.0e^{-5}$	$8.6e^{-3}$	$1.4e^{-2}$	$2.5e^{-3}$	$2.4e^{-3}$	$3.8e^{-3}$	$7.0e^{-4}$
Case 3	$1.2e^{-3}$	$6.3e^{-4}$	\	$2.0e^{-1}$	$2.4e^{-1}$	\	$5.5e^{-2}$	$6.6e^{-2}$	\
Case 4	$1.2e^{-3}$	$6.3e^{-4}$	\	$3.0e^{-1}$	$3.8e^{-1}$	\	$8.0e^{-2}$	$1.1e^{-1}$	\

CoM state. However, due to the divergence of the floating base estimation, the position of the CoM will also diverge over time.

5. Conclusions

Based on the humanoid robot nonlinear dynamics and DLKF, a nonlinear state estimation framework for humanoid robots was proposed, and we applied the framework to estimate the CoM and the DCM state.

In previous research, since the linear simplified models (LIPM or CIP) are too simple, many key dynamics in the humanoid robot locomotion are ignored, which makes it hard to obtain good prediction results at the prediction step. This article directly adopts the nonlinear model based on the CoM dynamics, without making any artificial assumptions about the dynamics of the robot, so that more accurate results are obtained in the prediction step. However, the nonlinear CoM dynamic model is also a simplified model compared to whole body dynamics as humanoid robots are highly complex dynamic systems. The simplification will inevitably introduce non-zero-mean, non-Gaussian and correlated errors in the prediction step. The traditional EKF cannot handle such errors well. This article proposed a CoM state estimator that combines DLKF with nonlinear dynamics. Since DLKF is robust to non-zero-mean, non-Gaussian, and correlated errors, the estimator well compensates for the process error caused by the parameter error, unmodeled drive system, and calibration error in the prediction step.

The virtual environment simulation results prove that compared with the traditional humanoid robots' linear and nonlinear state estimators, the nonlinear state estimation framework proposed in this paper shows better accuracy in the CoM and DCM state estimation. This is essential for the performance of the controllers that generally adopt the CoM or DCM as the feedback state.

We effectively fused the information from IMU, encoders, FSRs and CAD to estimate the state of humanoid robots. However, in the floating base kinematics estimator, the base position and yaw component of base orientation are unobservable, which makes the CoM position unobservable, and the error will grow over time. Thus, future work includes solving the unobservability with other sensors such as GPS, lidar and machine vision. Moreover, the contact detection mechanism described in Section 3.1 is very simplistic. Some more advanced contact detection methods [37,38] will be used in the future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

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%% Floating base kinematics estimation

% Contact Detector
    Current_Contact_foot = Contact_Detector(Sensor_FSR(left), Sensor_FSR(right));

% Estimate floating base kinematics
    Esti_Base_Pos = Base_Estimation_IEKF(Sensor_IMU_Gro, Sensor_IMU_Acc, Sensor_Encoder, CAD_Parameter);

%% CoM state estimation

% Forward kinematics
    Mea_CoM_Pos = Forward_Kinematics_CoM(Esti_Base_Pos, Sensor_Encoder, CAD_Parameter);
    Mea_ZMP_Pos = Forward_Kinematics_ZMP(Esti_Base_Pos, Sensor_Encoder, CAD_Parameter, Sensor_FSR);

% Caculate CoM acceleration
    Mea_CoM_Acc = CoM_acceleration_Caculator(Esti_Base_Pos, Mea_CoM_Pos, Sensor_IMU_Gro,
                  Sensor_IMU_Acc, Sensor_Encoder);

% Caculate angular momentum
    Mea_Ang_Mom = Ang_Mom_Caculator(Sensor_IMU_Gro, CAD_Parameter);

% Estimate CoM state
    [Esti_CoM_Pos, Esti_CoM_Vel, Esti_Ext_Froce] = CoM_Estimation_DLKF(Mea_CoM_Acc, Mea_Ang_Mom,
                                                   Mea_CoM_Pos, Mea_ZMP_Pos, Sensor_FSR, CAD_Parameter);
```

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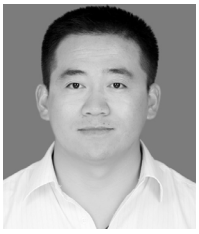
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